Frontier: Emerging-Term-Aware Search for the Latest AI Research Papers

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Code: `https://github.com/Nithurshen/frontier`

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Abstract

The vocabulary of AI research changes every few months. *GRPO*, *RLVR*, *MCP*, *vibe coding* and *agent skills* did not exist when today's off-the-shelf embedding models were trained. Literature search systems must still handle them. We present **Frontier**, a search engine over 302,070 arXiv papers in AI, ML, NLP and IR published between January 2023 and October 2026. Frontier combines a from-scratch inverted index (TF-IDF, BM25, query likelihood, pseudo-relevance feedback, Boolean retrieval, dynamic indexing) with dense retrieval. Its main novelty is an *emerging-term layer*. It detects new terms with Kleinberg's burst detection, a contrast against an older background corpus and a termhood filter. It links every term to the paper that introduced it, resolving homonyms by context similarity, and uses the links for query expansion and query-adaptive fusion. We also build **Fresh-2026**, a time-aware benchmark of 426 citation-context queries mined from 2026 arXiv papers that we crawled politely. On its held-out test split Frontier reaches nDCG@10 47.8, against 45.0 for a tuned BM25+dense hybrid and 40.9 for BM25. On queries that mention an emerging term it lifts the hybrid from 27.4 to 40.0. On the LitSearch benchmark (EMNLP 2024) our hybrid of BM25 and a 33M-parameter embedder reaches recall@5 64.1 on specific questions, above the published E5-large-v2 (56.2) and Instructor-XL (52.3) results.

1 Introduction

Background. Scientific literature search is a core information need. Its scale has exploded: our 2023–2026 slice of arXiv alone contains 302,070 papers in four AI categories, and the monthly volume roughly doubled over the period. Modern paper search engines rely on dense neural retrievers, which match queries and papers by meaning rather than exact words [2, 3].

Motivation. Dense retrievers are trained once and then frozen, while the field keeps inventing vocabulary. A model trained in 2023 has never seen *GRPO* (introduced in DeepSeekMath, 2024), the *Model Context Protocol* (2024–25) or *vibe coding* (2025). Such a model can only guess at their meaning from sub-word pieces. This is the temporal generalisation gap observed for language models [9], and the CLEF LongEval-Sci 2026 track reports terminology drift hurting dense retrieval of scientific papers [10]. Keyword search, on the other hand, sees the new token but knows nothing about its meaning, and a popular new term matches thousands of papers. A useful system for *latest* research must therefore know which terms are new, what they refer to, and which paper introduced them.

Importance. Researchers, students and reviewers mostly search for recent work. The newest papers are also the ones with no citations yet, so link-based ranking cannot help them [13]. Improving retrieval for new vocabulary directly improves access to the research front.

Contributions.

1. A complete search engine for recent AI papers, with every classic IR component built from scratch: inverted index with zone weighting, TF-IDF, BM25, query likelihood, pseudo-relevance feedback, Boolean retrieval, dynamic indexing, compression statistics, a polite crawler and a Streamlit demo.
2. An *emerging-term detector* that combines Kleinberg burst detection, background-corpus contrast and a termhood filter, together with *origin linking* that resolves homonyms such as MCP through context similarity.
3. *Emerging-term-aware ranking*: query expansion from the introducing paper, fusion weights that adapt to the query, and an origin boost.
4. **Fresh-2026**, a time-aware benchmark built from citation contexts in 146 papers from September–October 2026, plus a reproduction of LitSearch compared against its published results.

2 Related Work

Scientific literature retrieval. SPECTER [3] learns paper embeddings from citations. LitSearch [2] provides 597 expert-checked literature-search questions, built partly by rewriting *inline citation* sentences into questions. It shows that instruction-tuned dense retrievers such as GritLM-7B beat BM25 by about 25 points of recall@5. Context-aware citation recommendation [6] retrieves the paper cited at a given sentence, which is the task we use for Fresh-2026. BEIR [17] shows that BM25 remains a strong zero-shot baseline that dense models often fail to beat out of domain.

Agentic paper search. PaSa [5] trains an LLM agent that issues search queries and expands citations, and reports large recall gains over Google Scholar. SPAR [15] adds query decomposition and evolution with multiple agents. SAGE [7] benchmarks deep-research agents on scientific retrieval and finds that all of them struggle. It improves off-the-shelf retrievers by having an LLM add keywords and metadata to every document in the corpus. These systems need many LLM calls per query. Frontier instead adds a cheap, corpus-statistics layer that any retriever can use.

Temporal change. Lazaridou et al. [9] show that neural language models degrade on text written after their training period. LongEval-Sci [10] evaluates scientific retrieval as collections evolve and finds temporal signals and lexical matching more robust than a frozen dense model.

Emerging topics and terms. Kleinberg's burst model [8] detects periods where a term occurs at an elevated rate. Mei and Zhai [12] mine evolving themes in text streams. Contrast with a reference corpus (the *weirdness* measure [1]) is a classic way to separate domain terms from general language.

Retrieval models and fusion. We build on BM25 [14], Dirichlet-smoothed query likelihood [20], Rocchio feedback [11], rank fusion [4] and compact embedders (BGE [19], E5 [18]).

Novelty. To our knowledge, no prior search system (i) detects emerging terms online from corpus statistics, (ii) links each term to the paper that introduced it, using context to resolve homonyms, and (iii) uses the term to change *how* a query is ranked, through expansion, an adaptive lexical/dense balance and origin boosting. Prior work either retrains the retriever or calls an LLM per document or query [7, 5]. Fresh-2026 is also, to our knowledge, the first *time-aware* inline-citation benchmark for papers published after the training period of common embedders.

3 Objective

Given a natural-language description of a research idea, rank the arXiv papers published *before* the time of the query so that the paper the user means appears at the top. The aim is to do this robustly when the query uses vocabulary that appeared after the retrieval models were trained. Concretely, we aim to: (1) build a complete, efficient search engine for recent AI papers; (2) detect emerging terms and their introducing papers automatically; (3) improve retrieval for queries that contain such terms without hurting the others; (4) evaluate rigorously against published results and with significance tests.

4 Proposed Model

Figure 1 shows the pipeline. *Offline*, the arXiv snapshot and the papers crawled each day are tokenised into an inverted index and embedded by two dense encoders. The emerging-term layer scans monthly document frequencies, detects bursty new terms, filters them against the background corpus and links each to its origin paper, giving a *glossary*. *Online*, a query is matched against the glossary. Lexical (BM25) and dense scores are computed for the expanded query, fused with weights that depend on whether an emerging term was found, and the origin papers are boosted. The crawler adds new papers to an auxiliary index that is periodically merged.

Table 1: Inverted index statistics.

Statistic	arXiv	LitSearch
Documents	302,070	64,183
Dictionary terms	234,984	85,463
Postings	26,545,143	3,507,667
Tokens (title x2)	42,285,192	5,732,367
Average document length	139.98	89.31
Postings, 32-bit ids (MB)	101.26	13.38
Postings, VB gaps (MB)	28.84	4.0
Postings, γ gaps (MB)	25.46	3.89
Zipf slope	-1.691	-1.633

5 Methodology

5.1 Indexing

Titles and abstracts are lower-cased and split into alphanumeric tokens. Stop words are removed and tokens are Porter-stemmed (with a cache). A title occurrence counts twice, which gives a simple zone weighting. Postings are stored as a compressed sparse row matrix (terms × documents) holding term frequencies; this is the dictionary + postings structure with document ids in increasing order. Index statistics are in Table 1. Storing the gaps between document ids with variable-byte or γ codes would shrink the postings about 3.5–4×, and the collection follows Zipf's law (Fig. 6).

Dynamic indexing. Papers crawled each day go into an auxiliary index that shares the dictionary (new terms are appended). Queries score both indexes with global statistics (N, df). Once the auxiliary index exceeds 5,000 documents it is merged into the main index. A content-seen check (id and SHA-1 fingerprint of the normalised text) skips duplicates.

5.2 Retrieval models

For query q and document d with term frequency $\mathrm{tf}_{t,d}$, document length $|d|$ and average length $\overline{|d|}$:

BM25 [14]:

$$s(q,d) = \sum_{t \in q} \mathrm{idf}_t \, \frac{\mathrm{tf}_{t,d}\,(k_1+1)}{\mathrm{tf}_{t,d} + k_1\big(1 - b + b\frac{|d|}{\overline{|d|}}\big)}, \qquad (1)$$

with $\mathrm{idf}_t = \log\big(1 + \frac{N - \mathrm{df}_t + 0.5}{\mathrm{df}_t + 0.5}\big)$. k_1 and b are tuned on the dev split ($k_1 = 1.5$, $b = 0.9$).

TF-IDF (SMART lnc.ltc): documents get $1 + \log \mathrm{tf}$ weights with cosine normalisation, queries get $(1 + \log \mathrm{tf}) \cdot \log(N/\mathrm{df})$, and the score is their dot product.

Query likelihood with Dirichlet smoothing [20] ($\mu = 1000$): $s(q,d) = \sum_{t \in q} \log \frac{\mathrm{tf}_{t,d} + \mu P(t|C)}{|d| + \mu}$.

Pseudo-relevance feedback. We take the top 10 BM25 documents, add the 20 terms with the highest

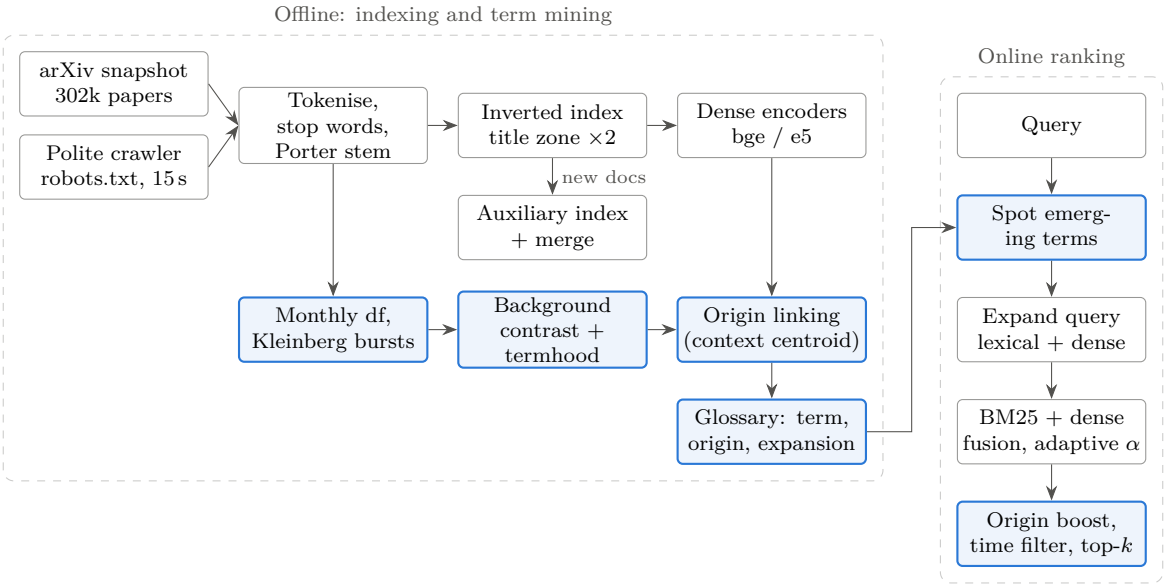


Figure 1: Frontier pipeline. Blue boxes are the new emerging-term components.

average normalised tf · idf to the query with weight 0.5 (Rocchio with $\gamma = 0$), and re-score.

Dense retrieval. bge-small-en-v1.5 [19] and e5-small-v2 [18] (33M parameters each, both trained in 2022–23) embed "title. abstract". The score is the cosine similarity with the query embedding.

Hybrid. Each model returns its top 1,000 documents. Over the union P, the scores are min-max normalised and combined:

$$s(d) = \alpha\, \hat{s}_{\mathrm{BM25}}(d) + (1-\alpha)\, \hat{s}_{\mathrm{dense}}(d). \qquad (2)$$

Boolean retrieval supports AND, OR, NOT, parentheses and quoted phrases, which are matched as conjunctions because the index stores no positions (parsed with the shunting-yard algorithm, with postings intersected in order of increasing length). **PageRank** [13] over the LitSearch citation graph (345k edges) gives a static quality prior $g(d)$.

5.3 Emerging-term detection

Candidates. We take all 1–3-grams of titles and abstracts that contain no stop word and are not pure numbers. For each month m we count the papers that contain the gram, r_m, out of d_m papers that month.

Growth and contrast. A candidate needs (a) at least 30 papers overall and 20 in the last six months, (b) a recent rate at least 8 times its rate in the first 18 months, and (c) a recent rate at least 50 times its rate in the older background corpus (64k LitSearch papers, 2017–2024). Condition (c) applies the weirdness idea [1] over time. Without it, the top "emerging terms" were *every*, *alone* and *strongest*: a 2026 shift in how abstracts are written, not new concepts.

Kleinberg bursts. For each surviving candidate we fit the two-state burst automaton [8]. The base state emits at $p_0 = \sum r_m / \sum d_m$ and the burst state at $p_1 = 3p_0$. The emission cost of state i in month m is

$$c_i(m) = -\big[r_m \log p_i + (d_m - r_m)\log(1-p_i)\big], \qquad (3)$$

and entering the burst state costs $\gamma \ln n$ ($\gamma = 1$, n months). The cheapest state sequence is found by Viterbi dynamic programming. A term is *emerging* if its last burst starts in July 2024 or later and its rate before the burst is under a quarter of its rate during it. Terms are ranked by burst weight $\sum_{m \in \mathrm{burst}} (c_0(m) - c_1(m))$.

Termhood. We keep terms that contain capitals or digits after the first character (GRPO, Qwen3-8B). We drop single lower-case words that are common in general English (word frequency ≥ 2.5 on the Zipf scale) and phrases with adverbs. The remaining lower-case terms must appear in at least 5% of the titles that use them. Overlapping n-grams with similar counts are merged, keeping the longer one ("model context protocol" rather than "context protocol"). We keep the top 1500.

5.4 Origin linking (term disambiguation)

The earliest paper that "introduces" an acronym is often a different concept altogether. MCP first appears in our corpus as a 2023 *Mass-Conserving Perceptron*, and DAPO and LLaDA have similar homonyms. We treat linking as entity disambiguation. Let c be the normalised mean bge embedding of up to 400 papers that use the term during its burst. Candidates are early papers that define the term (it is in the title, or in a sentence with "we propose/introduce/present"). We link to the *earliest* candidate d with $\cos(d, c) \geq \max_{d'} \cos(d', c) - 0.065$. The glossary stores, per term, the origin paper, a definition sentence and the ten highest-weighted stems of the origin paper as expansion terms.

5.5 Emerging-term-aware ranking

Given query q, we spot glossary terms by matching n-grams (longest match first), giving the set $E(q)$. (1) *Lexical expansion*: the BM25 query becomes $q' = q + \beta \sum_{e \in E(q)} \sum_w \frac{x_{e,w}}{\max x}\, w$, using the expansion weights x.

(2) *Dense expansion*: the text "e: title of the origin paper" is appended to the query before encoding. (3) *Adaptive fusion*: α becomes α_E when $E(q) \neq \emptyset$. (4) *Origin boost*: the origin papers are added to the pool and get $+\lambda$. On LitSearch we optionally add $\mu\, g(d)$ from PageRank. The parameters $(\alpha, \alpha_E, \beta, \lambda, k_1, b)$ are grid-searched on the dev split to maximise nDCG@10. The selected values were $\alpha = 0.5$, $\alpha_E = 0.5$, $\beta = 0.1$, $\lambda = 0.75$, $k_1 = 1.5$ and $b = 0.9$. The origin boost (4) is applied only to *named* terms (containing capitals or digits after the first character), since generic concepts such as "verifier" have no single introducing paper. Size variants (Qwen3-4B) inherit the origin of their model family (Qwen3), and a candidate whose title *starts* with the term ("Qwen3 Technical Report", "DAPO: …") is preferred.

5.6 Fresh-2026 benchmark construction

We sampled 420 papers from September–October 2026 with a primary category in cs.CL, cs.IR, cs.LG or cs.AI, and fetched their arXiv HTML. robots.txt allows `/html` and asks for a 15 s crawl delay, which we respect. For each citation in the body text we resolve the bibliography entry to a corpus paper, through an arXiv identifier or an exact match of the normalised title. A sentence becomes a query if it cites one or two corpus papers that are all resolvable and older than the citing paper, and has 8–60 words without math, code or figure references. The citation markers are removed. We keep at most four queries per paper. The cited papers are the relevant documents. Retrieval is *time-aware*: only papers published before the citing paper may be returned, which simulates searching at that date. Queries are split by citing paper into dev and test halves, so no paper contributes to both. Each query is labelled by the age of its cited papers: *old* (before July 2024), *new* (after July 2025) or *mixed*. The glossary is built only from papers before September 2026, so it never sees the citing papers.

5.7 Evaluation metrics

For a query with relevant set R and ranking $d_1, d_2, \ldots$: recall $R@k = |R \cap \{d_1..d_k\}|/|R|$; precision $P@k = |R \cap \{d_1..d_k\}|/k$; nDCG@10 $= \frac{1}{Z}\sum_{i=1}^{10} \frac{[d_i \in R]}{\log_2(i+1)}$; MRR $=$ 1/rank of the first relevant; AP $= \frac{1}{|R|}\sum_{i:d_i \in R} P@i$, averaged as MAP. We also plot 11-point interpolated precision–recall curves. Term detection is scored as a classification with precision, recall, F1 and accuracy against manual labels. Significance uses paired two-sided t-tests and Wilcoxon signed-rank tests over queries [16].

6 Experimentation and Results

6.1 Datasets

arXiv corpus. The 27 October 2026 arXiv metadata snapshot, filtered to papers listed in cs.CL, cs.LG, cs.AI or cs.IR and first submitted between January 2023 and October 2026: 302,070 titles and abstracts and 234,984 index terms. **Fresh-2026.** We requested 201 sampled papers before arXiv began throttling the crawler. 186 had an HTML version, and 146 produced at least one valid query, giving 426 queries: 210 for tuning (dev) and 216 for testing. 47 test queries contain an emerging term. **LitSearch** [2]: 597 queries (inline-citation and author-written, specific and broad) over 64,183 ACL/ICLR papers. We use the published titles and abstracts and report recall@5 for specific and recall@20 for broad questions, as the paper does. **Term labels.** 230 terms were labelled as genuine new technical terms or not: the top 60 detected terms, a random sample of 70 other detected terms, 60 rejected high-growth candidates, and a reference list of 40 well-known 2024–2026 terms. The origin links of 37 named artefacts (methods, models, benchmarks, tools) were also checked.

Table 2: Fresh-2026 test split (216 queries, time-aware). $+E$ marks emerging-term expansion. Best in bold.

System	R@5	R@20	R@100	nDCG@10	MRR	MAP
TF-IDF	41.4	53.2	62.3	42.1	42.6	39.5
QL-Dirichlet	40.5	50.5	59.5	39.2	39.2	36.4
BM25	42.4	50.9	59.5	40.9	41.2	38.2
BM25+PRF	42.1	50.7	58.8	39.4	39.3	36.4
BM25+E	43.1	51.4	60.0	41.4	41.7	38.7
e5-small	34.3	44.9	53.0	31.1	31.3	28.4
e5-small+E	34.7	45.4	55.6	32.0	32.4	29.2
bge-small	39.1	48.4	58.6	36.3	36.9	33.0
bge-small+E	39.1	49.5	59.3	36.7	37.4	33.6
Hybrid	46.3	57.4	69.4	45.0	46.1	42.9
Hybrid+E	47.7	57.9	69.9	45.8	46.4	43.4
Frontier	**49.5**	**60.0**	**70.8**	**47.8**	**48.5**	**45.1**

Setup. All parameters are tuned on the Fresh-2026 dev split and then fixed; LitSearch is never used for tuning. Everything runs on one laptop (Apple M4 Pro, 48 GB). Mean latency per query over 302,070 papers is 12.3 ms for BM25 and 47.3 ms for Frontier, including query encoding on the GPU (Fig. 7).

6.2 Main results on Fresh-2026

Table 2 lists all systems on the test split. Frontier is best on every metric: nDCG@10 47.8, against 45.0 for the tuned hybrid, 42.1 for TF-IDF, 40.9 for BM25 and 36.3 for bge-small. The improvement over the hybrid is significant ($p_t = 0.013$; better on 14 queries, worse on 3). Fusion is the largest step among the baselines: it adds 4 points over BM25 and 9 over bge-small, because lexical and dense retrievers fail on different queries. The small 2023 embedders are weaker than BM25 on their own on this domain, as BEIR [17] also found out of domain. Table 5 gives all paired tests.

6.3 Effect of emerging terms

Table 3 and Fig. 3 split the test queries by whether they contain an emerging term.

(1) All baselines struggle on new vocabulary. BM25 drops from 46.7 to 19.9 nDCG@10. Emerging terms are not rare tokens: GRPO appears in 1,800 papers and Qwen3 in hundreds, so they get a low IDF and match many papers that merely *use* the term. The cited paper, however, is usually the one that *introduced* it. The dense models also lose 10–18 points.

(2) Expansion helps every retriever: BM25 19.9 → 22.2, bge-small 28.6 → 30.5, e5-small 16.9 → 21.3, hybrid 27.4 → 31.1. The gains are consistent, but with

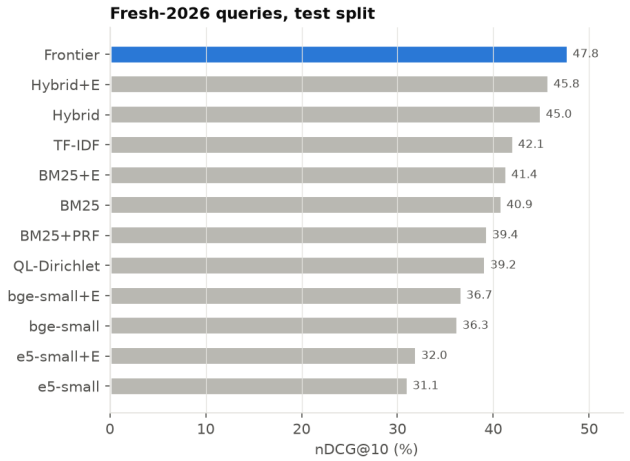


Figure 2: nDCG@10 on the Fresh-2026 test split.

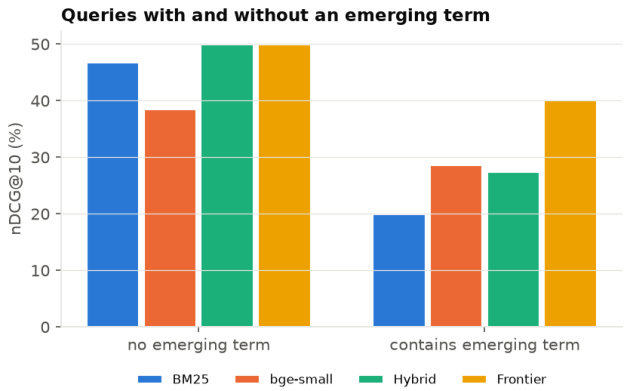


Figure 3: nDCG@10 by query type: every baseline drops on new vocabulary; Frontier recovers most of the gap.

Table 3: nDCG@10 on test queries with and without an emerging term.

System	no emerging term	emerging term
BM25	46.7	19.9
bge-small	38.4	28.6
e5-small	35.0	16.9
Hybrid	**49.9**	27.4
Hybrid+E	**49.9**	31.1
Frontier	**49.9**	**40.0**

Table 4: nDCG@10 by publication date of the cited paper (test split).

System	before 2024-07	mixed	after 2025-07
BM25	34.8	34.1	59.2
bge-small	29.8	32.7	50.4
e5-small	26.5	24.1	47.8
Hybrid	37.4	38.9	64.6
Hybrid+E	37.7	41.2	64.1
Frontier	**40.4**	**42.7**	**65.4**

47 queries none is significant on its own.

(3) Origin linking gives the largest gain: Frontier reaches 40.0 on these queries, **+12.6 points over the hybrid** ($p_t = 0.011$, better on 14 queries, worse on 3). Linking a named term to its introducing paper is exactly what an inline citation needs.

(4) No harm elsewhere: on queries without emerging terms Frontier equals the hybrid (49.9 vs. 49.9), because the layer only switches on when a term is found.

Age of the cited paper. Table 4 splits queries by when the cited paper appeared. Every system finds queries about *new* papers (after July 2025) easier, because recent work is usually cited by a distinctive name. The gap between BM25 and the 2023 dense model, however, is widest exactly there: 59.2 vs. 50.4 for new papers, against 34.8 vs. 29.8 for papers from before July 2024. This fits the vocabulary drift that motivates our work. Frontier is best in every age group.

6.4 Comparison with a published paper: LitSearch

Table 6 compares our systems with the results published in LitSearch [2], using the same queries, corpus and metrics. Our BM25 (stemming, title zone, k_1, b tuned on Fresh-2026) reaches 55.8 recall@5 on specific questions, against 50.0 published. Our hybrid of BM25 and the 33M-parameter bge-small reaches 64.1 (broad recall@20 60.6). This beats the published E5-large-v2 (56.2/55.4) and Instructor-XL (52.3/56.5), and narrows the gap to GritLM-7B (74.8/70.8), a model about 200 times larger. LitSearch's papers predate our emerging terms: only 11 of its 597 queries contain one, and no origin links exist in that corpus. Frontier therefore behaves exactly like the hybrid there (64.1), which shows the new layer is safe on older literature. The PageRank prior leaves recall@5 unchanged but gives a small, significant nDCG@10 gain (51.5 → 51.9, $p_t = 0.006$). Citation authority helps break ties between candidates.

6.5 Emerging-term detection and linking

Table 7 scores term detection as a classification against the labels. Precision is high among the top-ranked terms (0.61) and lower in the long tail (0.22 on a random sample of the rest). The detector recovers 0.65 of the reference terms and correctly rejects 0.98 of the high-growth generic candidates. The missed reference terms are mostly older ones whose burst began before July 2024, such as *test-time scaling* and *LLM-as-a-judge*. Origin linking is correct for 17 of the 37 named artefacts, i.e. 77% of those whose introducing paper is on arXiv. Examples: GRPO → DeepSeekMath, Qwen3-4B → Qwen3 Technical Report, DAPO, GSPO, GEPA, DeepSeek-R1/V3, SWE-bench, BrowseComp, HarmBench, SGLang and FineWeb-Edu. Product names announced outside arXiv (GPT-5.x, Claude Code, MCP) are linked to an early paper about them instead. Context-based disambiguation is essential here. Without it, MCP was linked to a 2023 *Mass-Conserving Perceptron*, DAPO to an unrelated *Dual Approximation Policy Optimization*, and LLaDA to a driving paper. Fig. 5 shows detected bursts.

Table 5: Paired significance tests on the Fresh-2026 test split. W/L = queries where A is better / worse than B; p_t paired t-test, p_W Wilcoxon signed-rank.

Comparison	Metric	Mean A	Mean B	W/L	p_t	p_W
Frontier vs Hybrid	nDCG@10	47.8	45.0	14/3	0.013	0.010
Frontier vs Hybrid	R@20	60.0	57.4	6/1	0.027	0.019
Frontier vs BM25	nDCG@10	47.8	40.9	43/19	2e-5	3e-5
Frontier vs BM25	R@20	60.0	50.9	24/3	2e-5	3e-5
Frontier vs bge-small	nDCG@10	47.8	36.3	72/15	3e-8	3e-8
Frontier vs bge-small	R@20	60.0	48.4	39/8	4e-6	1e-5
Hybrid+E vs Hybrid	nDCG@10	45.9	45.0	9/3	0.177	0.147
Hybrid+E vs Hybrid	R@20	57.9	57.4	2/2	0.528	0.458
BM25+E vs BM25	nDCG@10	41.4	40.9	5/1	0.144	0.075
BM25+E vs BM25	R@20	51.4	50.9	1/0	0.318	0.317
bge-small+E vs bge-small	nDCG@10	36.7	36.3	8/7	0.616	0.570
bge-small+E vs bge-small	R@20	49.5	48.4	5/3	0.298	0.248
Hybrid vs BM25	nDCG@10	45.0	40.9	35/19	0.001	0.002
Hybrid vs BM25	R@20	57.4	50.9	20/4	9e-4	0.001

Table 6: LitSearch: our systems (top) against the published results of Ajith et al. [2] (bottom). Recall@5 for specific and recall@20 for broad questions, titles and abstracts only.

System	Inline-citation		Author-written		Average	
	Spec. R@5	Broad R@20	Spec. R@5	Broad R@20	Spec. R@5	Broad R@20
TF-IDF	43.5	43.3	60.2	51.4	51.5	45.1
QL-Dirichlet	43.7	41.4	61.1	60.0	52.0	45.6
BM25	46.5	44.6	65.9	57.1	55.8	47.5
BM25+PRF	41.8	45.9	63.0	57.1	51.9	48.4
e5-small	43.7	38.9	53.1	54.3	48.2	42.4
bge-small	51.9	59.4	55.5	57.1	53.6	58.9
Hybrid	55.6	58.3	73.5	68.6	64.1	60.6
Frontier	55.6	58.3	73.5	68.6	64.1	60.6
Frontier+PageRank	56.5	58.3	72.5	65.7	64.1	60.0
BM25 [2]	38.5	37.4	62.6	48.6	50.0	39.9
GTR-T5-large [2]	38.5	45.7	40.8	37.1	39.6	43.8
Instructor-XL [2]	48.9	56.3	55.9	57.1	52.3	56.5
E5-large-v2 [2]	50.4	55.8	62.6	54.3	56.2	55.4
GritLM-7B [2]	67.7	69.7	82.5	74.3	74.8	70.8

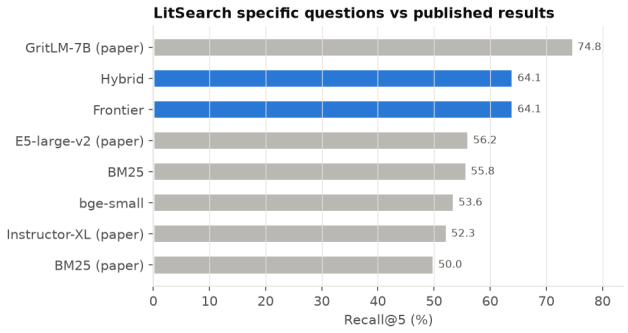


Figure 4: LitSearch specific questions: ours (blue) vs. published retrievers.

Table 7: Emerging-term detection against manual labels.

Set	n	P	R	F1	Acc.
Top-60 detected	60	0.61	–	–	–
Random detected	70	0.22	–	–	–
Reference list	40	–	0.65	–	–
Rejected	60	–	–	–	0.98
Combined	169	0.43	0.71	0.54	0.59

6.6 Discussion

Three lessons stand out. First, classic IR does most of the work. A tuned BM25 is a strong baseline, and a simple min-max fusion with a small embedder beats much larger dense models on LitSearch. Second, new vocabulary is a real weakness of *both* retriever families, but for different reasons. Lexical matching cannot tell which of the many papers using a popular new term is meant. Dense models trained before the term existed do not know what it means. Third, the most effective fix is not richer matching but *knowledge*: knowing which paper introduced a term. Tuning on the dev split chose a strong origin boost ($\lambda = 0.75$) with light expansion ($\beta = 0.1$). It kept the same fusion weight with or without an emerging term ($\alpha_E = \alpha = 0.5$), so the adaptive-fusion component turned out to be unnecessary once expansion and linking were in place. The gains cost nothing at query time: Frontier's latency equals the hybrid's (47.4 ms).

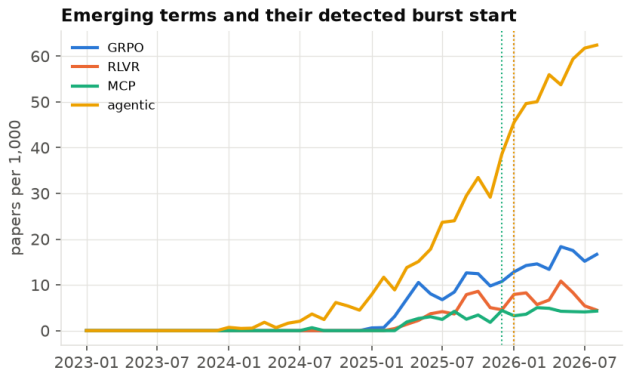


Figure 5: Monthly rate of four emerging terms; dotted lines mark the start of the latest burst found by Kleinberg's automaton.

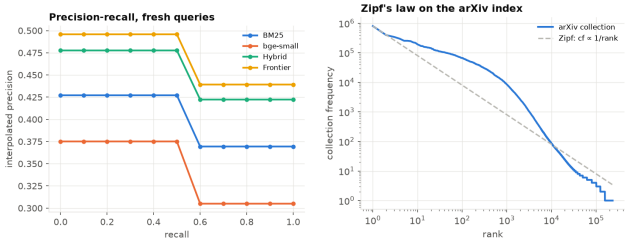


Figure 6: Left: interpolated precision–recall on Fresh-2026. Right: Zipf's law on the arXiv index.

7 Conclusions and Limitations

We built Frontier, a complete search engine for the latest AI research. Detecting emerging terms from corpus statistics, and linking them to the papers that introduced them, improves retrieval significantly. It adds +12.6 nDCG@10 on queries with new vocabulary and +2.8 overall over a tuned hybrid, without retraining any model or calling an LLM at query time. Our hybrid baseline also beats several much larger published retrievers on LitSearch.

Limitations. (1) Fresh-2026 is built automatically: a citation sentence is an imperfect query, an author may cite one of several valid papers, and unresolved citations are dropped. With 216 test queries (47 with emerging terms), small differences are not significant. (2) The title-prefix and model-family rules for origin linking were added after inspecting linking errors on Fresh-2026 queries from both splits. The test-split gain of the origin boost may therefore be slightly optimistic. (3) Term detection is noisy in the long tail (precision 0.22). The labels come from the project itself, not from independent annotators. (4) Origins cannot be found for terms introduced outside arXiv (blogs, product launches). (5) We use small 2023 embedders because of laptop compute. Newer models (e.g. Qwen3-Embedding) would raise all dense numbers, but their own training cutoff moves the vocabulary gap rather than closing it. (6) We index titles and abstracts only. arXiv HTML crawling at the required 15 s delay, plus throttling, limited Fresh-2026 to 146 citing papers. **Future work**: extracting definitions for glossary entries without an LLM, term-aware document expansion at index time, and a continuously refreshed benchmark.

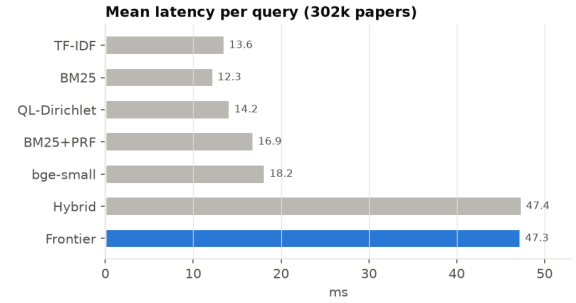


Figure 7: Mean latency per query over 302,070 papers.

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